Stage 2 – Completing Implementation & Research

After finishing Stage 1 (frontend and basic ARCore floor detection with map adding), we moved to Stage 2. Here we did a research survey comparing indoor navigation methods, implemented manual ARCore navigation with checkpoints, manual step counting using gyroscope and speedometer for distance/direction, created research paper and PPT, and submitted to conference.

Week 1 – Firebase Auth Setup

In Week 1 of Stage 2, we completed Firebase authentication fully. We enabled email/password sign-in in the Firebase console and added signup, login, and forgot password logic implementation in our Android app. Users now get an email link to reset passwords if they forget, making the app secure and user-friendly.

We tested auth flows end-to-end: new users register with email and password, store user data in Firebase, and login smoothly. We handled errors like weak passwords or invalid emails. By week's end, all users must log in to access station selection and AR navigation.

Week 2 – Manual ARCore Arrows with Checkpoints

Week 2 focused on manual ARCore navigation implementation. We created a manual map, added checkpoints from source to destination, and rendered stable AR arrows based on straight paths and number of turns between checkpoints.

We fixed arrow flickering using ARCore pose tracking and smoothing. Arrows show direction at specific distances with manual turn counts. We tested in indoor spaces simulating station halls.

Developer mode was polished – we scanned, placed, and saved multiple anchors with POI (Point of Interest) details manually. By end of week, end-to-end worked: select points → manual checkpoints → follow AR arrows.

Week 3 – Research Survey on Indoor Navigation Methods

In Week 3, we conducted literature survey comparing indoor positioning: WiFi (4-8m accuracy but needs infrastructure), BLE (3-5m but beacon deployment), vs IMU sensors (accelerometer/gyroscope step counting 93-97% accuracy, no extra hardware).

We studied 20+ papers on ARCore navigation, manual step detection via peak analysis on accelerometer magnitude, and gyroscope for heading/turns. IMU methods showed best for real-time without setup costs.

This survey justified our choice: ARCore + manual IMU step counting over radio-based methods. We documented comparisons with accuracy tables for paper.

Week 4 – Manual Step Counting Implementation

Week 4 implemented manual step counting using phone sensors. We used accelerometer for step detection (peak detection on magnitude data) and gyroscope for direction/turns, estimating distance with stride length.

Speedometer data helped refine distance per step. Manual checkpoints reset drift – count steps from source to next checkpoint, show AR arrows for straight/turn paths.

Tests showed 95%+ step accuracy across phone positions. Integrated with ARCore: steps update position along manual map path.

Week 5 – Full Sensor Integration & Testing

Week 5 fused gyroscope (turns/heading) + speedometer (distance) with ARCore. Manual path: source → checkpoints → destination with predefined step counts and turn directions.

AR arrows trigger at checkpoint distances: "straight 50 steps" or "turn left after 20 steps". Tested full flow in 100m indoor path – minimal drift with manual resets.

Validated against survey: our IMU+ARCore (2-3m error) beat WiFi/BLE infrastructure needs. Ready for research documentation.

Week 6 – App Integration & Demo Preparation

Week 6 joined everything: login → station select → manual map with checkpoints → start AR navigation showing step count, distance remaining, turn alerts via arrows.

Fixed checkpoint arrow snapping and added "recalibrate at checkpoint" using ARCore anchors. Recorded full demo video: user follows manual steps/turns to destination.

App stable across devices. Prepared data/logs for paper: step accuracy graphs, path error vs survey methods.

Week 7 – Research Paper, PPT & Conference Submission

Week 7 wrote research paper "ARCore+IMU Step Counting for Railway Station Navigation". Included survey (WiFi/BLE/IMU comparisons), our manual method (checkpoints, sensor fusion), results (95% step accuracy, real-time turns), future scope.

Created PPT with architecture, demos, comparison tables. Submitted to student conference, now waiting for acceptance response and presentation slot.

Stage 2 complete – working app, validated research, ready for conference.